COURSE 1 · MATHEMATICS FOUNDATIONS

Mathematics foundations

Printable workbook with worked solutions

56 problems across 7 modules, plus the course capstone.
Full worked solutions at the back.

Name _____ Started _____

Before you start

How to use this workbook

There are 56 problems, eight per module, followed by the course capstone. Dots show difficulty: one dot is a warm-up, two is the standard you are aiming for, three stretches towards first-year university.

Work in pencil and show your steps. Try each problem before looking at the solutions, which start on the page headed *Worked solutions*. The same problems are in the interactive workbook in the Second Pass app, which checks answers instantly. The app also has a quiz for every module and course, and spaced-repetition flashcards; a printable set of the flashcards comes with each course.

> **A good routine for each module**
>
> Problems 1–2 straight after listening to the reader. Problems 3–6 after the explorer. Problems 7–8 at the end of the week, then take the module quiz in the app and redo anything you got wrong without looking. Review the module's flashcards for a few minutes each day after that.

Module 1.1

Numbers and arithmetic fluency

1.1.1 ●○○

A jacket priced at $180 is reduced by 35%. What is the sale price?

Answer: _____ dollars

1.1.2 ●○○

A town's population grows from 24 000 to 27 600. What is the percentage increase?

Answer: _____ %

1.1.3 ●●○

Support for a policy rises from 40% to 50%. Which statement is correct?

A. It rose by 10%

B. It rose by 10 percentage points, or 25%

C. It rose by 25 percentage points

D. It rose by 50%

1.1.4 ●●○

Calculate $(6 \times 10^{5}) \times (4 \times 10^{-2})$. Give the answer as an ordinary number.

Answer: _____

1.1.5 ●●○

Light travels at 3.0×10^{8} m/s. The Sun is 1.5×10^{11} m away. How many seconds does sunlight take to reach Earth?

Answer: _____ s

1.1.6 ●●○

A red blood cell is about 8 micrometres across. How many would fit side by side across 1 centimetre?

Answer: _____

1.1.7 ●●○

A calculator gives 3.14159265 for a result whose inputs were measured to 3 significant figures. How should you report it?

A. 3.14159265

B. 3.142

C. 3.14

D. 3

1.1.8 ●●●

Estimate, to the nearest power of ten, how many times your heart beats in an 80-year life. Show your assumptions.

Module 1.2

Units, measurement and dimensional analysis

1.2.1 ●○○

Convert 90 km/h into metres per second.

Answer: _____ m/s

1.2.2 ●○○

A room is 4.0 m by 3.5 m. What is its floor area in square centimetres?

Answer: _____ cm^2

1.2.3 ●●○

A tank holds 2.5 cubic metres of water. How many litres is that?

Answer: _____ L

1.2.4 ●●○

Which of these equations is dimensionally impossible? (v speed, t time, a acceleration, s distance)

A. $s = vt$

B. $v = at$

C. $s = v + at$

D. $v^2 = 2as$

1.2.5 ●●○

What are the SI base units of a joule?

A. kg m s^{-1}

B. kg m^2 s^{-2}

C. kg m s^{-2}

D. N s

1.2.6 ●●○

A length is measured as 25.0 ± 0.5 cm. What is the percentage uncertainty?

Answer: _____ %

1.2.7 ●●●

A rectangle's sides are 20.0 cm ± 1% and 10.0 cm ± 2%. What is the percentage uncertainty in its area?

Answer: _____ %

1.2.8 ●●●

A friend averages 50 readings from a scale that always reads 0.2 kg too high. Explain why the average is still wrong, and what kind of error this is.

Module 1.3

Geometry

1.3.1 ●○○

A ladder 5.0 m long leans against a wall with its foot 1.4 m from the wall. How high up the wall does it reach?

Answer: _____ m

1.3.2 ●○○

What is the area of a circle of diameter 10 cm?

Answer: _____ cm^2

1.3.3 ●●○

A box measures 30 cm × 40 cm × 120 cm. How long is the longest straight rod that fits inside?

Answer: _____ cm

1.3.4 ●●○

A model car is built at 1:20 scale. Its paintwork covers 0.15 m^2. What area of paint covers the real car?

Answer: _____ m^2

1.3.5 ●●○

If you double the radius of a sphere, by what factor does its volume increase?

Answer: _____ ×

1.3.6 ●●○

Earth's radius is about 6.37×10^6 m. Treating it as a sphere, what is its volume?

Answer: _____ m^3

1.3.7 ●●○

A cube's side length triples. Its surface-area-to-volume ratio

A. triples

B. stays the same

C. falls to one third

D. falls to one ninth

1.3.8 ●●●

Use the square-cube law to explain why elephants have proportionally thicker legs than gazelles.

Module 1.4

Trigonometry

1.4.1 ●○○

A ramp rises at 12° and is 4.0 m long along its surface. How high does it rise?

Answer: _____ m

1.4.2 ●○○

Convert 45° to radians, as a decimal.

Answer: _____ rad

1.4.3 ●●○

From 50 m away, the top of a tower is at an angle of elevation of 35°. How tall is the tower (ignoring eye height)?

Answer: _____ m

1.4.4 ●●○

A rope pulls a sled with 200 N at 25° above the horizontal. What is the horizontal component of the pull?

Answer: _____ N

1.4.5 ●●○

A wheel of radius 0.30 m turns through 2.5 radians. How far does a point on the rim travel?

Answer: _____ m

1.4.6 ●●○

A right triangle has opposite side 7 and hypotenuse 10. Find the angle, in degrees.

Answer: _____ °

1.4.7 ●●○

The wave $y = 3\sin(2t)$ has amplitude and period

A. 3 and 2

B. 3 and π

C. 2 and 3

D. 6 and π

1.4.8 ●●●

Explain, using the unit circle, why $\sin^2\theta + \cos^2\theta = 1$ for every angle.

Module 1.5

Probability and statistics

1.5.1 ●○○

A fair coin is tossed three times. What is the probability of three heads? Give a decimal.

Answer: _____

1.5.2 ●○○

Two dice are rolled. What is the probability that the total is 7? Give a decimal.

Answer: _____

1.5.3 ●●○

Find the mean of these timings in seconds: 2.1, 2.3, 1.9, 2.2, 2.0.

Answer: _____ s

1.5.4 ●●○

A set of measurements has standard deviation 0.40 s. You take 16 readings. What is the standard error of the mean?

Answer: _____ s

1.5.5 ●●○

Heights are normally distributed with mean 170 cm and SD 10 cm. Roughly what fraction are between 150 cm and 190 cm?

A. 68%

B. 95%

C. 99.7%

D. 50%

1.5.6 ●●○

In a class, 60% study physics and, of those, 30% also study chemistry. What fraction of the class studies both? Give a decimal.

Answer: _____

1.5.7 ●●●

A disease affects 1 in 1000 people. A test detects it 99% of the time and gives false positives 1% of the time. If someone tests positive, what is the probability they have the disease? Give a decimal.

Answer: _____

1.5.8 ●●●

To halve the standard error of a mean, how many more readings do you need? Explain.

Module 1.6

Vectors

1.6.1 ●○○

Find the magnitude of the vector (6, 8).

Answer: _____

1.6.2 ●○○

Which of these is a vector?

A. temperature

B. mass

C. velocity

D. energy

1.6.3 ●●○

A walker goes 3 km east then 4 km north. What is the size of her displacement?

Answer: _____ km

1.6.4 ●●○

Add (2, −5) and (7, 1). What is the magnitude of the result?

Answer: _____

1.6.5 ●●○

A plane flies at 250 km/h on a heading 30° north of east. What is its northward velocity component?

Answer: _____ km/h

1.6.6 ●●○

Calculate the dot product of $(3, -2, 4)$ and $(1, 5, 2)$.

Answer: _____

1.6.7 ●●●

Find the angle between $(1, 0)$ and $(1, 1)$, in degrees, using the dot product.

Answer: _____ °

1.6.8 ●●●

Two vectors have a dot product of zero, and neither is the zero vector. What can you conclude, and why?

Module 1.7

Mathematical thinking

1.7.1 ●○○

Someone claims 'all prime numbers are odd'. What settles it?

A. Checking primes up to 100

B. The counter-example 2

C. A survey of mathematicians

D. It cannot be settled

1.7.2 ●○○

Use the formula $n(n+1)/2$ to find $1 + 2 + \ldots + 100$.

Answer: _____

1.7.3 ●●○

Prove that the product of two odd numbers is odd.

1.7.4 ●●○

A model of a falling stone ignores air resistance. For which object does this assumption fail most badly?

A. a steel ball dropped 2 m

B. a feather

C. a brick dropped 1 m

D. a marble

1.7.5 ●●○

Exponential growth predicts a bacterial colony doubling forever. Which observation shows the model has broken down?

A. Early growth is fast

B. Growth slows as nutrients run out

C. Cells divide by fission

D. The colony is visible

1.7.6 ●●○

A student derives that the time for a ball to fall a height h is $t = \sqrt{(2h/g)}$. Check the formula in the limit $h \to 0$ and by units.

1.7.7 ●●●

In a proof by contradiction, you

A. prove the statement for one example

B. assume the statement is false and derive an impossibility

C. check all cases by computer

D. assume what you want to prove

1.7.8 ●●●

Explain why a scientific theory can be overturned but a mathematical theorem cannot.

Capstone

Estimating Earth from a shadow

Around 240 BCE Eratosthenes noticed that at noon on the summer solstice the Sun shone straight down a well at Syene, while at Alexandria, about 800 km to the north, a vertical stick cast a shadow. Rebuild his estimate of Earth's size using geometry, trigonometry, units and uncertainty from this course.

Part 1 ●●○

At Alexandria, a stick 1.00 m tall casts a shadow 0.126 m long. What angle do the Sun's rays make with the vertical, in degrees?

Answer: _____ °

Part 2 ●●○

That angle equals the angle at Earth's centre between the two cities. What fraction of a full circle is it? Give a decimal.

Answer: _____

Part 3 ●●○

The cities are 800 km apart. Estimate Earth's circumference in km.

Answer: _____ km

Part 4 ●●○

From that circumference, estimate Earth's radius in km.

Answer: _____ km

Part 5 ●●●

If the 800 km distance is uncertain by ±5% and the angle by ±3%, what is the approximate percentage uncertainty in the circumference?

Answer: _____ %

Part 6 ●●●

List two assumptions in Eratosthenes' method and say how each could bias the answer.

Answers

Worked solutions

Module 1.1 · Numbers and arithmetic fluency

1.1.1 ANSWER 117 dollars

1 35% of 180 is 0.35 × 180 = 63.

2 180 − 63 = 117. Or directly: 0.65 × 180 = 117.

1.1.2 ANSWER 15 %

1 Change = 3600.

2 3600 ÷ 24 000 = 0.15, which is 15%.

1.1.3 ANSWER B. It rose by 10 percentage points, or 25%

1 The difference is 10 percentage points.

2 Relative to the starting 40, that is 10 ÷ 40 = 25%.

1.1.4 ANSWER 2.4e+04

1 Coefficients: 6 × 4 = 24. Exponents: 5 + (−2) = 3.

2 $24 \times 10^3 = 2.4 \times 10^4 = 24\,000$.

1.1.5 ANSWER 500 s

1 time = distance ÷ speed.

2 1.5 ÷ 3.0 = 0.5 and $10^{11} \div 10^8 = 10^3$.

3 $0.5 \times 10^3 = 500$ s, a little over eight minutes.

1.1.6 ANSWER 1250

1 8 µm = 8×10^{-6} m. 1 cm = 10^{-2} m.

2 $10^{-2} \div (8 \times 10^{-6}) = 0.125 \times 10^4 = 1250$.

1.1.7 ANSWER C. 3.14

1 Match the least precise input: 3 significant figures.

2 3.14.

1.1.8 ANSWER About 70 beats a minute × 60 minutes × 24 hours × 365 days × 80 years. Rounding: $70 \times 60 \approx 4 \times 10^3$ per hour; × 24 ≈ 10^5 per day; × 365 ≈ 3.7×10^7 per year; × 80 ≈ 3×10^9. So roughly three billion, of order 10^9.

1 Choose a rate per minute, then convert step by step to a lifetime.

2 Round aggressively; the aim is the power of ten.

Module 1.2 · Units, measurement and dimensional analysis

1.2.1 ANSWER 25 m/s

1 Multiply by 1000 m/km and divide by 3600 s/h.

2 90 000 ÷ 3600 = 25 m/s.

1.2.2 ANSWER 1.4×10^5 cm^2

1 Area = 14 m^2.

2 1 m^2 = 10^4 cm^2, so $14 \times 10^4 = 1.4 \times 10^5$ cm^2.

1.2.3 ANSWER 2500 L

1 1 m^3 = 1000 L.

2 2.5 × 1000 = 2500 L.

1.2.4 ANSWER C. $s = v + at$

1 Check each term. v has units m/s; at has units m/s; s has units m.

2 $s = v + at$ adds metres to metres-per-second: impossible.

1.2.5 ANSWER B. kg m^2 s^{-2}

1 Joule = newton × metre.

2 N = kg m s^{-2}, so J = kg m^2 s^{-2}.

1.2.6 ANSWER 2 %

1 0.5 ÷ 25.0 = 0.02.

2 0.02 × 100 = 2%.

1.2.7 ANSWER 3 %

1 For multiplication, add percentage uncertainties.

2 1% + 2% = 3%.

1.2.8 ANSWER This is a systematic error: every reading is shifted by the same 0.2 kg in the same direction. Averaging cancels random scatter but not a constant offset, so the mean is still 0.2 kg too high. The fix is to calibrate the scale, for example by weighing a known mass, and correct for the offset.

1 Ask: does the error push every reading the same way?

Module 1.3 · Geometry

1.3.1 ANSWER 4.8 m

1 The ladder is the hypotenuse: $h^2 + 1.4^2 = 5.0^2$.

2 $h^2 = 25 - 1.96 = 23.04$, so h = 4.8 m.

1.3.2 ANSWER 78.54 cm^2

1 Radius = 5 cm.

2 $A = \pi \times 5^2 \approx 78.5$ cm^2.

1.3.3 ANSWER 130 cm

1 $d = \sqrt{(30^2 + 40^2 + 120^2)}$.

2 $= \sqrt{(900 + 1600 + 14\,400)} = \sqrt{16\,900} = 130$ cm.

1.3.4 ANSWER 60 m^2

1 Areas scale by $k^2 = 20^2 = 400$.

2 $0.15 \times 400 = 60$ m^2.

1.3.5 ANSWER 8 ×

1 $V \propto r^3$.

2 $2^3 = 8$.

1.3.6 ANSWER 1.08×10^{21} m^3

1 $V = (4/3)\pi r^3$.

2 $r^3 = 2.58 \times 10^{20}$; × 4.19 gives about 1.08×10^{21} m^3.

1.3.7 ANSWER C. falls to one third

1 Area scales by 9, volume by 27.

2 Ratio scales by 9/27 = 1/3.

1.3.8 ANSWER Weight goes with volume, so it scales with length cubed. The strength of a leg bone goes with its cross-sectional area, which scales with length squared. A larger animal built to the same proportions would carry far more weight per unit of bone area, so the bones would fail. To compensate, large animals need disproportionately thick legs.

1 Which quantity goes with volume, and which with area?

Module 1.4 · Trigonometry

1.4.1 ANSWER 0.8316 m

1 Height is opposite the angle; ramp is the hypotenuse.

2 $h = 4.0 \times \sin 12° \approx 0.83$ m.

1.4.2 ANSWER 0.7854 rad

1 Multiply by π/180.

2 45π/180 = π/4 ≈ 0.785 rad.

1.4.3 ANSWER 35.01 m

1 tan 35° = height ÷ 50.

2 height = 50 × tan 35° ≈ 35 m.

1.4.4 ANSWER 181.3 N

1 Horizontal is adjacent to the angle.

2 200 × cos 25° ≈ 181 N.

1.4.5 ANSWER 0.75 m

1 s = rθ with θ in radians.

2 0.30 × 2.5 = 0.75 m.

1.4.6 ANSWER 44.43 °

1 sin θ = 7/10 = 0.7.

2 $\theta = \sin^{-1}(0.7) \approx 44.4°$.

1.4.7 ANSWER B. 3 and π

1 Amplitude is the coefficient: 3.

2 Period = 2π/ω = 2π/2 = π.

1.4.8 ANSWER On the unit circle, the point at angle θ has coordinates (cos θ, sin θ). Its distance from the centre is always the radius, 1. By Pythagoras, the horizontal and vertical distances satisfy $\cos^2\theta + \sin^2\theta = 1^2 = 1$, whatever θ is.

1 What are the coordinates of the point, and how far is it from the origin?

Module 1.5 · Probability and statistics

1.5.1 ANSWER 0.125

1 Independent: multiply.

2 ½ × ½ × ½ = 1/8 = 0.125.

1.5.2 ANSWER 0.1667

1 Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) = 6.

2 6 ÷ 36 = 1/6 ≈ 0.167.

1.5.3 ANSWER 2.1 s

1 Sum = 10.5.

2 10.5 ÷ 5 = 2.1 s.

1.5.4 ANSWER 0.1 s

1 SE = s/√n.

2 0.40 ÷ √16 = 0.40 ÷ 4 = 0.10 s.

1.5.5 ANSWER B. 95%

1 150 and 190 are two SDs either side of the mean.

2 About 95%.

1.5.6 ANSWER 0.18

1 P(both) = P(physics) × P(chemistry | physics).

2 0.6 × 0.3 = 0.18.

1.5.7 ANSWER 0.09016

1 Imagine 100 000 people: 100 have it, 99 test positive.

2 Of 99 900 without it, 1% = 999 test positive.

3 P = 99 ÷ (99 + 999) ≈ 0.090, about 9%.

1.5.8 ANSWER Four times as many. The standard error is s/√n, so it shrinks with the square root of n. To divide it by 2 you must multiply √n by 2, which means multiplying n by 4. Precision improves slowly with more data.

1 SE depends on 1/√n.

Module 1.6 · Vectors

1.6.1 ANSWER 10

1 $\sqrt{(6^2 + 8^2)} = \sqrt{100}$.

2 = 10.

1.6.2 ANSWER C. velocity

1 Only velocity needs a direction.

1.6.3 ANSWER 5 km

1 Components (3, 4).

2 √(9 + 16) = 5 km.

1.6.4 ANSWER 9.849

1 Sum = (9, −4).

2 √(81 + 16) = √97 ≈ 9.85.

1.6.5 ANSWER 125 km/h

1 North is opposite the 30° angle from east.

2 250 × sin 30° = 125 km/h.

1.6.6 ANSWER 1

1 3×1 + (−2)×5 + 4×2.

2 3 − 10 + 8 = 1.

1.6.7 ANSWER 45 °

1 a·b = 1. |a| = 1, |b| = √2.

2 cos θ = 1/√2, so θ = 45°.

1.6.8 ANSWER They are perpendicular. The dot product equals |a||b|cos θ. Neither magnitude is zero, so cos θ must be zero, which means θ = 90°.

1 Use the geometric form of the dot product.

Module 1.7 · Mathematical thinking

1.7.1 ANSWER B. The counter-example 2

1 One counter-example disproves a general claim.

2 2 is prime and even.

1.7.2 ANSWER 5050

1 n = 100.

2 100 × 101 ÷ 2 = 5050.

1.7.3 ANSWER Write the odd numbers as 2m + 1 and 2n + 1. Their product is 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1, which is one more than an even number, so it is odd.

1 Write a general odd number as 2m + 1.

1.7.4 ANSWER B. a feather

1 Air resistance matters when it is large compared with weight.

2 A feather is light with a large surface.

1.7.5 ANSWER B. Growth slows as nutrients run out

1 The model assumes unlimited resources.

2 Slowing growth signals that limit.

1.7.6 ANSWER If h = 0 then t = 0, which is right: no fall takes no time. Units: h in m, g in m/s^2, so h/g has units s^2, and its square root is s, a time. Both checks pass.

1 Substitute h = 0, then check units of h/g.

1.7.7 ANSWER B. assume the statement is false and derive an impossibility

1 That is the definition of the method.

1.7.8 ANSWER A theorem follows by logic from stated definitions and axioms, so it stays true as long as those assumptions hold; nothing observed in the world can change a deduction. A scientific theory makes claims about the world and is tested against evidence, so new observations can show it is wrong or limited, as Newton's mechanics was shown to fail at very high speeds.

1 Where does each get its authority: logic or evidence?

Capstone · Estimating Earth from a shadow

Part 1 ANSWER 7.181 °

1 tan θ = shadow ÷ height = 0.126.

2 $\theta = \tan^{-1}(0.126) \approx 7.2°$.

Part 2 ANSWER 0.01995

1 7.2 ÷ 360.

2 ≈ 0.020, or one fiftieth.

Part 3 ANSWER 4.01e+04 km

1 Arc = fraction × circumference.

2 800 ÷ 0.0199 ≈ 40 000 km.

Part 4 ANSWER 6383 km

1 $C = 2\pi r$, so $r = C/2\pi$.

2 40 000 ÷ 6.28 ≈ 6400 km.

Part 5 ANSWER 8 %

1 Circumference = distance ÷ fraction: a division.

2 Add percentage uncertainties: 5% + 3% = 8%.

Part 6 ANSWER (1) The Sun is so far away that its rays arrive parallel; if not, the angle would not equal the central angle. (2) Syene lies due south of Alexandria; if it is off to one side, the 800 km is not the true north–south arc, and the circumference would be overestimated. Others: Earth is a perfect sphere; the distance was measured accurately.

1 What has to be true for the shadow angle to equal the angle at Earth's centre?